Dendrite Segmentation in SEM Images

Classical Image Processing vs Deep Learning

Abstract

Lithium dendrites are microscopic metallic structures that form during battery charging and may lead to internal short circuits and thermal runaway. Automatic segmentation of dendrites in scanning electron microscopy (SEM) images can assist in analyzing their morphology and preventing battery failure.

This project investigates two different approaches for dendrite segmentation:

1. **A classical image processing pipeline based on morphological operations**
2. **A deep learning model based on YOLO segmentation**

The dataset consists of **14 annotated SEM images** containing dendritic structures.

Both methods are evaluated against manually annotated ground truth masks using the following segmentation metrics:

• **IoU (Intersection over Union)**
• **Dice coefficient**
• **Precision**
• **Recall**

Experimental results show that the YOLO segmentation model achieves slightly better overall performance than the classical pipeline, while both approaches produce competitive segmentation results.

1. Introduction

Lithium dendrites are metallic structures that grow on battery electrodes during repeated charging cycles. These structures can penetrate the separator between electrodes and cause internal short circuits, leading to **thermal runaway and battery failure**.

Scanning Electron Microscopy (SEM) is often used to analyze dendrite formation. However, **manual analysis of SEM images is time-consuming and subjective**. Automatic segmentation techniques can assist in identifying dendritic structures and analyzing their morphology.

This project compares two approaches for automatic dendrite segmentation:

• **Classical computer vision techniques**
• **Deep learning based segmentation using YOLO**

The goal is to evaluate the **strengths and limitations of each approach when applied to noisy SEM imagery**.

The main contributions of this project are:

• **Implementation of a classical segmentation pipeline for dendrite extraction in SEM images**

• **Training of a YOLO-based deep learning segmentation model**

• **Quantitative comparison between classical and deep learning approaches**

• **Visual analysis of segmentation successes and failure cases**

This comparison helps highlight the strengths and limitations of deterministic image processing pipelines versus data-driven deep learning models for noisy microscopic imagery.

2. Dataset

The dataset used in this project consists of **14 scanning electron microscopy (SEM) images** containing lithium dendritic structures.

The images were **manually annotated using Roboflow**, where dendrite regions were labeled using **polygon segmentation annotations**.

These annotations define the boundaries of dendrite structures within each image.

The dataset was exported in **YOLO segmentation format**, which stores **normalized polygon coordinates** for each annotated object.

Before annotation, the raw **SEM TIFF images** were converted to **uint8** using **percentile-based normalization** and cropped by **93 pixels** from the bottom to remove the fixed **SEM overlay**. The resulting **PNG images** were then used for **Roboflow annotation** and for preparation of the **YOLO segmentation dataset**.

For training the deep learning model, the dataset was split into three subsets:

- **Training set - 70% (10 images)**
- **Validation set - 20% (2 images)**
- **Test set - 10% (2 images)**

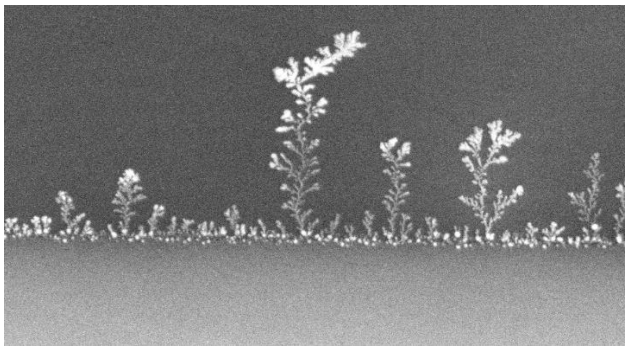
The images represent two levels of segmentation difficulty:

EASY - Relatively clean SEM images with clearly visible dendrite structures.

HARD - Noisy SEM images containing complex dendritic patterns and background artifacts.

This dataset split allows evaluation of model performance on **test images** that were not used during training, while monitoring training stability using the **validation set**.

Example SEM image containing lithium dendritic structures:



3. Methodology

Two different segmentation approaches were implemented to extract dendritic structures from SEM images.

The first approach is based on a **classical image processing pipeline** that applies filtering, adaptive thresholding, and morphological operations to isolate dendrite structures.

The second approach uses a **deep learning segmentation model based on YOLO**, which is trained on manually annotated SEM images to automatically predict dendrite masks.

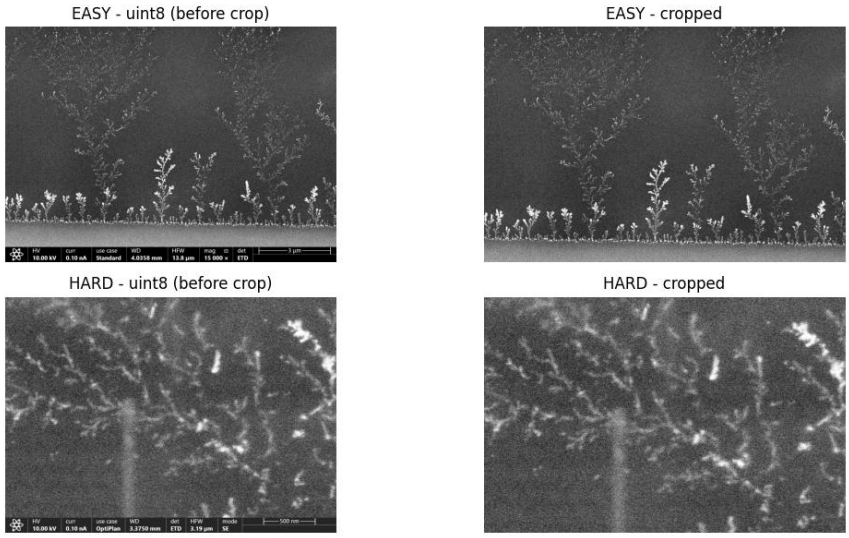
Both approaches are later evaluated and compared using **quantitative segmentation metrics**.

3.1 Classical Image Processing Pipeline

The classical segmentation pipeline consists of several processing stages designed to progressively isolate dendritic structures from SEM images.

Image Normalization and Overlay Removal

The raw SEM TIFF images may contain high dynamic range values and a fixed bottom overlay (scale bar and text).
To ensure consistent processing, the images were first converted to **uint8** using percentile-based contrast normalization.
In addition, a constant crop of **93 pixels from the bottom** was applied to remove the overlay region.



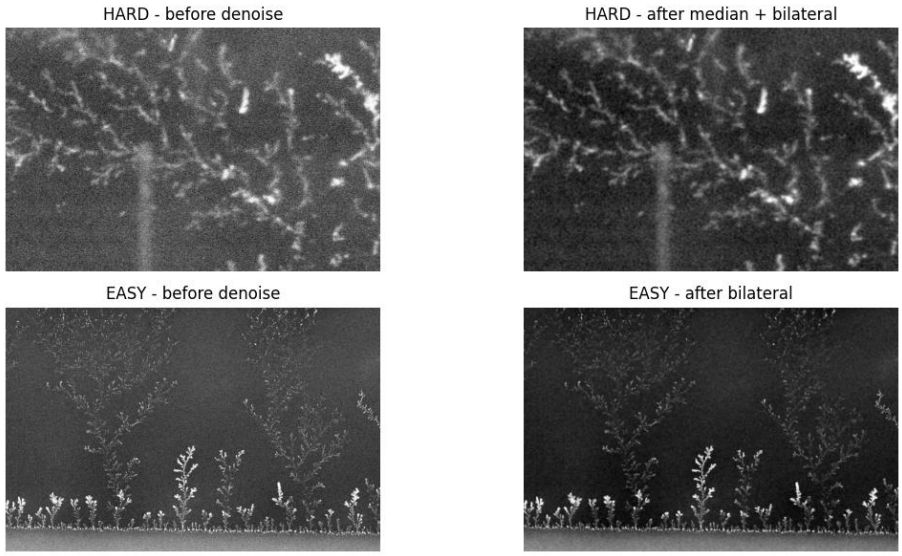
These images illustrate the preprocessing stage used to standardize the input data.

Edge-Preserving Denoising

SEM images often contain granular noise that may interfere with segmentation.
To reduce noise while preserving dendrite edges, an edge-preserving denoising procedure was applied.

For **EASY** images, **bilateral filtering** was used to smooth the image while preserving intensity edges, which is important for maintaining thin dendritic branches.

For **HARD** images, an additional **median filter (disk radius = 2)** was applied before the bilateral filter to suppress stronger granular noise patterns.



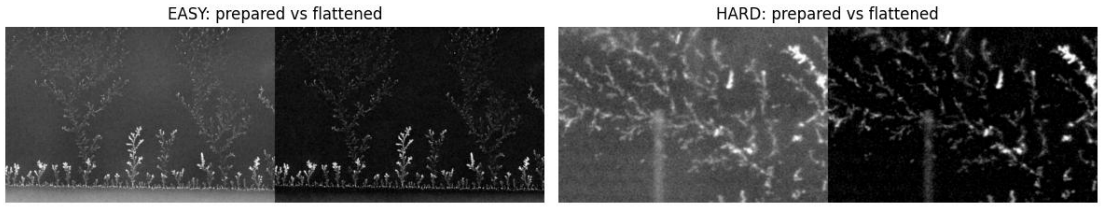
This step reduces noise while maintaining the structural boundaries of dendritic branches.

Background Flattening

SEM images may contain slow illumination gradients that affect thresholding stability.
To mitigate this effect, background flattening was applied before adaptive thresholding.

For EASY images, the background was estimated using **morphological opening with a large structuring element**.
For HARD images, **morphological reconstruction** was used to better handle strong illumination variations and noise.

The estimated background was subtracted from the denoised image and the result was normalized back to the range **0-255**.



This step improves the robustness of the thresholding stage.

Sauvola Adaptive Thresholding

Sauvola adaptive thresholding was selected because it performs well on images with uneven illumination and textured backgrounds, which are common in SEM microscopy.

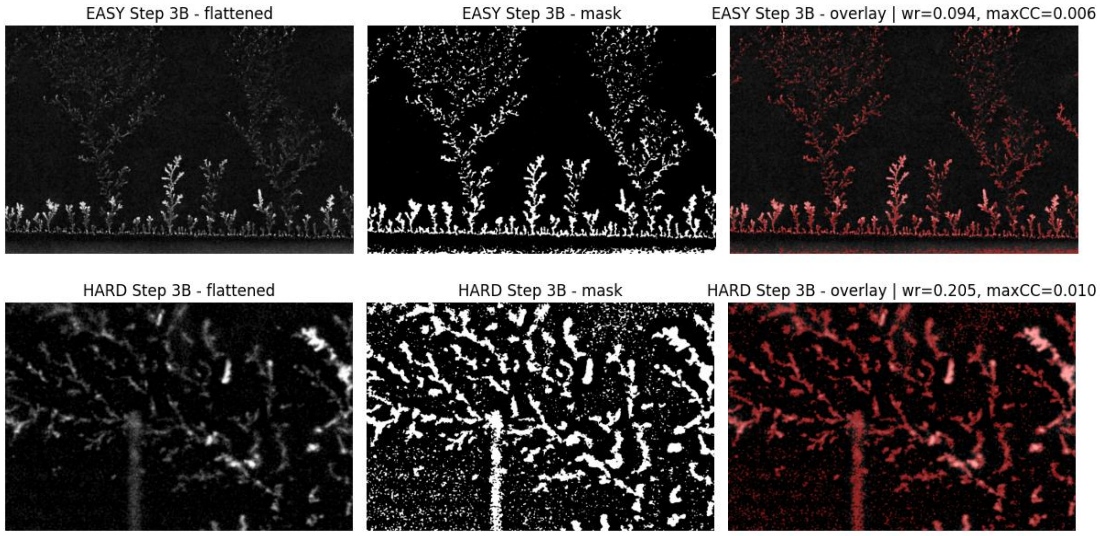
After background correction, adaptive thresholding was used to extract the initial dendrite mask.

The flattened image was first inverted to highlight bright dendrites on a dark background.

Then **Sauvola adaptive thresholding** was applied to generate a binary mask separating dendritic structures from the background.

The **Sauvola window size** was set to **101** for both subsets. The parameter **k** was set to **0.09** for **EASY** images and **0.04** for **HARD** images, reflecting the stronger noise conditions in the **HARD** subset.

For EASY images, a light **morphological closing** operation was applied to reconnect small gaps in thin branches.



These images illustrate how dendritic structures are initially separated from the background.

Morphological Mask Cleanup

The binary masks produced after Sauvola thresholding may still contain noise, small disconnected components, and artifacts near the bottom of the SEM images.
To refine the segmentation while preserving thin dendritic branches, a morphological cleanup stage was applied.

For **EASY images**, a conservative morphological cleanup strategy was used. The operations included:

• Removal of very small connected components using `remove_small_objects`
• Filling small holes inside dendritic regions using `remove_small_holes`
• Morphological closing to reconnect small gaps in thin dendritic branches
• Suppression of artifacts in the lower region of the image using bottom-band filtering

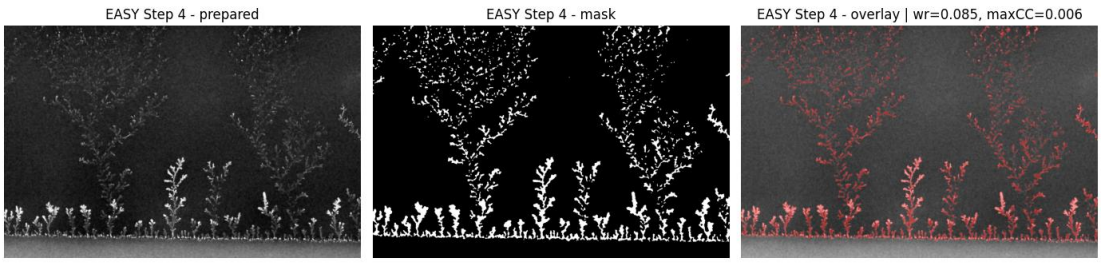
These operations remove isolated noise while preserving the connectivity of the dendritic network.

For **HARD images**, the segmentation masks contain significantly more speckle noise and fragmented components.
Therefore, a stronger refinement strategy was applied.

First, **morphological reconstruction** was used to suppress detached noise components while preserving structures connected to larger dendritic regions.
Additional rule-based filtering was then applied to remove weak or very small components and suppress artifacts near the bottom boundary.

These steps reduce noise while maintaining the main dendritic structures.

The final refined masks are illustrated below for both EASY and HARD examples.
Each visualization shows the prepared image, the cleaned binary mask, and an overlay of the mask on the original image.



This stage significantly improves the structural consistency of the segmentation masks.

Branch Separation Using Distance Transform and Watershed

After the morphological cleanup stage, some dendritic branches may still remain connected by thin bridges.
These narrow connections can merge neighboring structures and reduce the accuracy of the subsequent skeletonization and topology analysis.

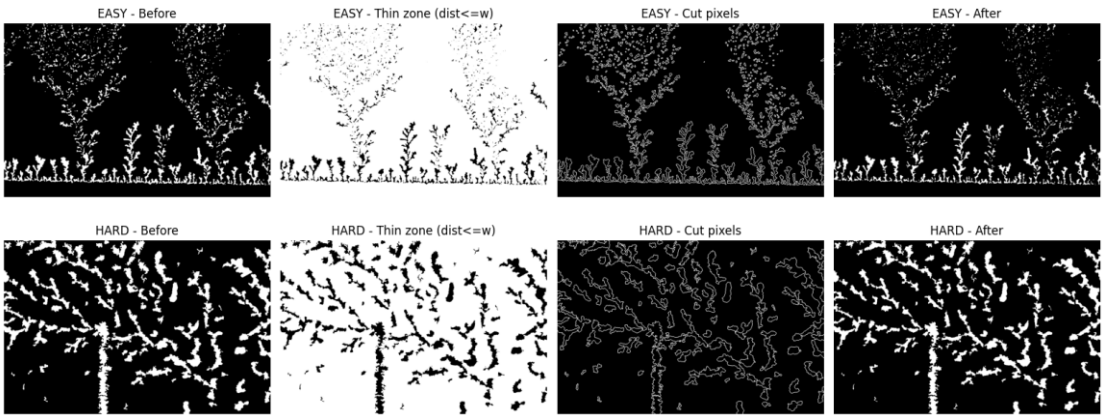
To address this issue, a branch separation step was applied using a **distance transform and watershed-based refinement**.

First, the **Euclidean distance transform** was computed on the cleaned binary mask.
This produces a distance map in which each pixel represents its distance from the nearest background pixel.

Thin regions were then identified using a distance threshold.
Watershed segmentation was applied to detect boundaries between adjacent dendritic regions.

Finally, boundary pixels were removed **only within these thin regions**, allowing narrow bridges between branches to be separated while preserving thicker dendritic structures.

This operation selectively separates connected dendritic branches without significantly altering the overall segmentation mask.



These visualizations demonstrate how thin artificial connections between dendritic branches are removed while maintaining the main dendritic structures.

The watershed-based branch separation modified only a small portion of the segmentation mask.
For EASY images, about **2.35% of the pixels** were changed, while for HARD images about **2.08%** were modified.

Although the change is limited, this step removes thin artificial bridges between dendritic branches and improves the structural separation before the skeletonization stage.

Skeletonization

In the final stage, the refined binary mask was converted into a **1-pixel-wide skeleton representation** using a skeletonization algorithm.

Skeletonization preserves the topology of the dendritic network while reducing each structure to its centerline.

This representation enables analysis of **dendrite morphology**, including **branch endpoints** and **branching points**.

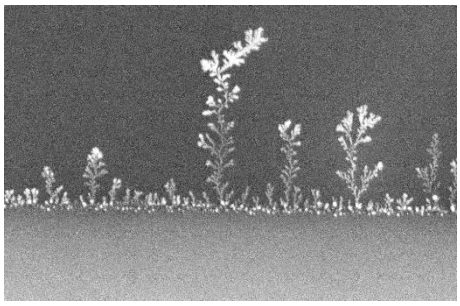
Note on CLAHE preprocessing

CLAHE was evaluated as a potential preprocessing step to enhance the contrast of thin dendritic branches. However, it also amplified the granular SEM background noise, which increased false positive detections during thresholding. Therefore, **CLAHE was not included in the final pipeline**.

Classical Pipeline Output

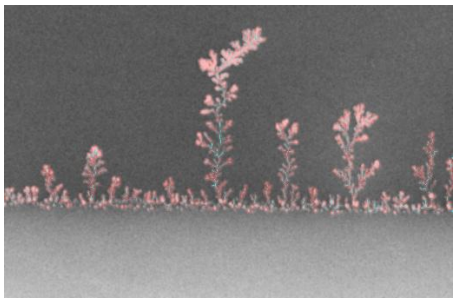
Output of the classical image processing pipeline showing the extracted dendrite mask, overlay visualization, and skeleton representation.

Original SEM Image



Classical Mask

Overlay Visualization



Skeleton Representation

3.2 Deep Learning Approach

In addition to the classical pipeline, a deep learning segmentation approach was trained to automatically detect dendritic structures in SEM images.

Two YOLO segmentation models were trained during the deep learning stage of the project.

The first model used the **YOLOv11n-seg** architecture, which is a lightweight segmentation model designed for efficient object detection and instance segmentation tasks. This initial model was trained as a baseline using the following parameters:

- **Epochs:** 50
- **Image size:** 640
- **Batch size:** 4

Although the YOLOv11n model was able to detect dendritic structures, its segmentation quality was limited by the relatively small model capacity and short training duration.

To improve performance, a second model based on **YOLOv11s-seg** was trained. This model has higher representational capacity and was trained for a longer duration using transfer learning with pretrained YOLO segmentation weights.

The final model used in this project is therefore: **YOLOv11s-seg**

The main training parameters of the final model were:

• **Epochs:** 150
• **Image size:** 640
• **Batch size:** 4

The number of training epochs was increased to 150 to allow the model to converge on the small dataset.

An **image resolution of 640** was selected because it preserves fine dendritic structures while remaining compatible with standard **YOLO segmentation settings**.

A **batch size of 4** was used due to GPU memory constraints and the relatively high resolution of the images.

During training, the model learns to predict **segmentation masks** for dendrite structures based on the annotated **polygon labels**.

After training, the model with the best **validation performance** was automatically saved as **best.pt**.

This final trained model was later used for the **inference stage**, where segmentation masks were generated for **unseen SEM images** and compared with the **classical image processing pipeline**.

During inference, the trained **YOLO segmentation model** was applied using the following parameters:

• **Confidence threshold:** 0.2
• **IoU threshold:** 0.5
• **Image size:** 640

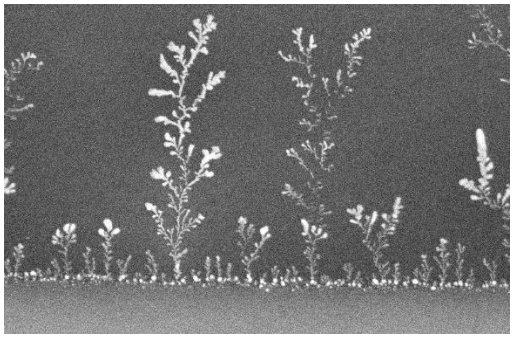
The **confidence threshold** controls the minimum probability required for a detection to be considered valid.

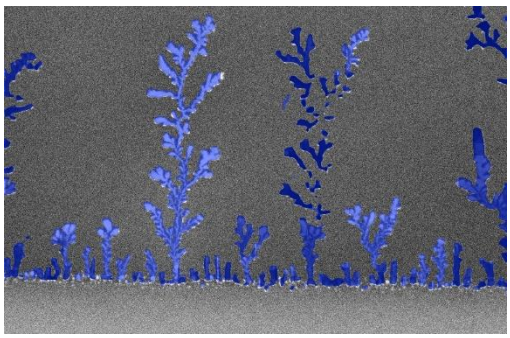
A relatively moderate threshold (**0.2**) was selected to allow detection of **thin dendritic branches** that may produce lower confidence predictions.

The **IoU threshold (0.5)** is used during **non-maximum suppression** to remove overlapping detections while preserving distinct dendritic structures.

The **image size of 640 pixels** was chosen because it preserves **fine dendritic details** while remaining consistent with the **training resolution**.

The following example illustrates a segmentation produced by the final trained **YOLO model**.





4. Results and Discussion

The performance of the classical image processing pipeline and the YOLO segmentation model was evaluated using several segmentation metrics.

The evaluation was performed by comparing the predicted masks with the manually annotated ground truth masks.

The following metrics were used:

• **IoU (Intersection over Union)** - measures the overlap between predicted and ground truth masks
• **Dice coefficient** - measures similarity between segmentation masks
• **Precision** - proportion of predicted dendrite pixels that are correct
• **Recall** - proportion of ground truth dendrite pixels that were successfully detected

Table 1: Average segmentation performance across the dataset for the classical pipeline and YOLO segmentation model using **IoU, Dice, Precision, and Recall.**

Method	IoU	Dice	Precision	Recall
Classical Pipeline	0.545	0.695	0.733	0.702
YOLO Segmentation	0.584	0.729	0.716	0.763

The quantitative results show that **both approaches achieve relatively strong segmentation performance** on the dataset.

The classical image processing pipeline achieves an average **IoU of 0.545** and **Dice score of 0.695**, demonstrating that the morphological processing pipeline is able to capture a substantial portion of the dendrite structures.

The YOLO segmentation model achieves slightly higher overlap with the ground truth annotations, reaching an **IoU of 0.584** and **Dice score of 0.729**.

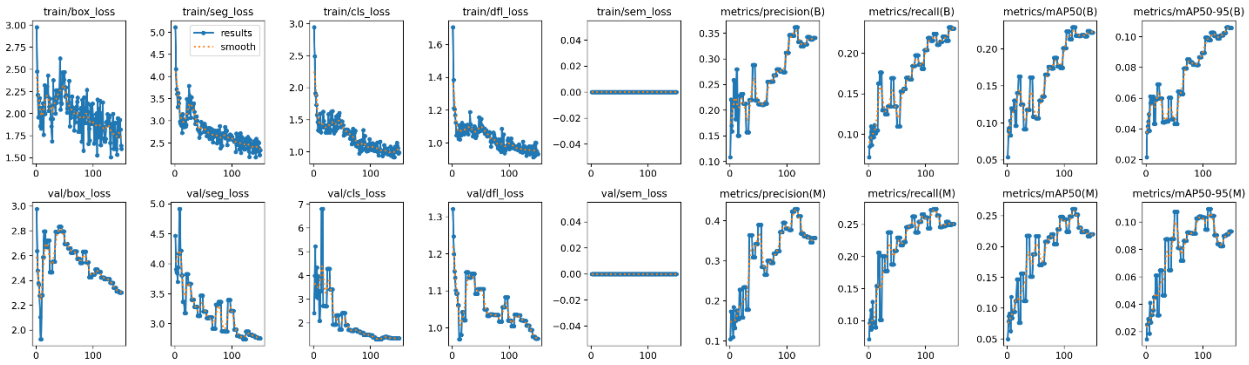
Interestingly, the two approaches demonstrate **different strengths**. The classical pipeline achieves **higher Precision (0.733)**, indicating that its predictions contain fewer false positive pixels. In contrast, the YOLO model achieves **higher Recall (0.763)**, meaning that it detects a larger portion of the true dendrite structures.

Overall, the deep learning model performs **slightly better on average**, but the classical pipeline remains **a strong baseline method that produces competitive results on this dataset**.

YOLO Training Curves

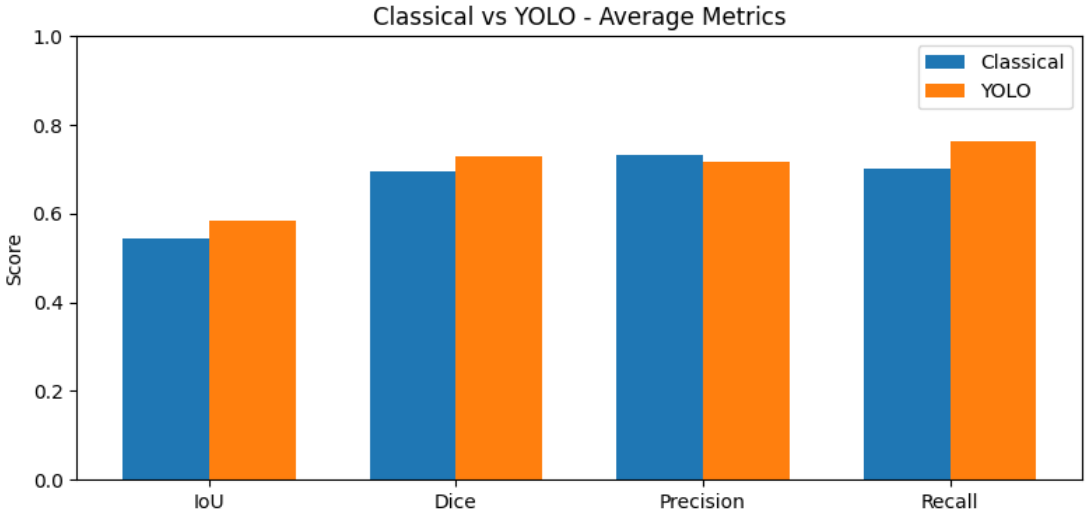
The following graph shows the training behavior of the final YOLO segmentation model over **150 epochs**.

The curves illustrate the evolution of the training losses and segmentation metrics during model training.



Average Metrics Comparison

The following graph presents a visual comparison between the classical image processing pipeline and the YOLO segmentation model using the four evaluation metrics: IoU, Dice, Precision, and Recall.



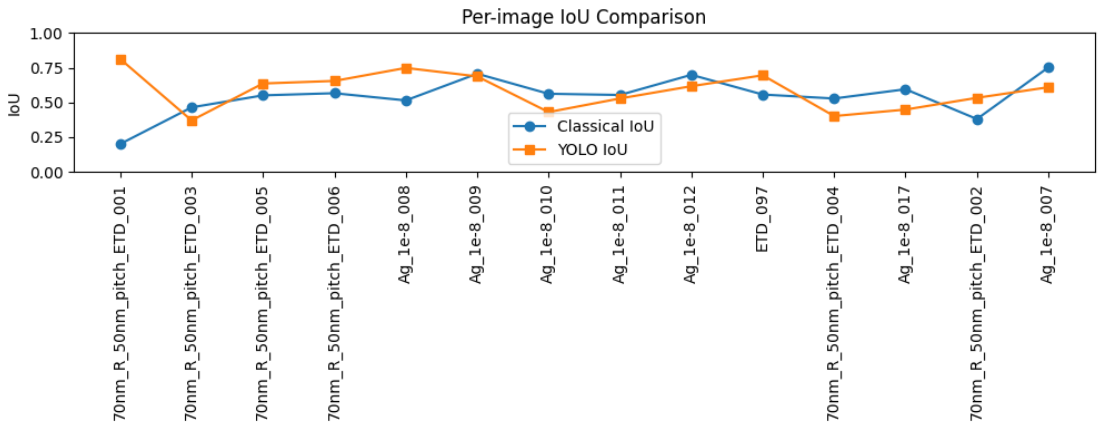
The graph shows that the **YOLO segmentation model achieves slightly higher IoU, Dice, and Recall values** compared to the classical pipeline.

However, the classical method achieves **higher Precision**, indicating that it produces more conservative segmentations with fewer false positive detections.

These results highlight that **both approaches perform competitively**, with different strengths in terms of detection sensitivity and prediction accuracy.

Per-image Performance

The following graph shows the IoU performance for each individual image in the dataset.



Overall, the YOLO segmentation model achieves **slightly higher IoU values in many images**, particularly in cases where dendrite structures are complex or fragmented.

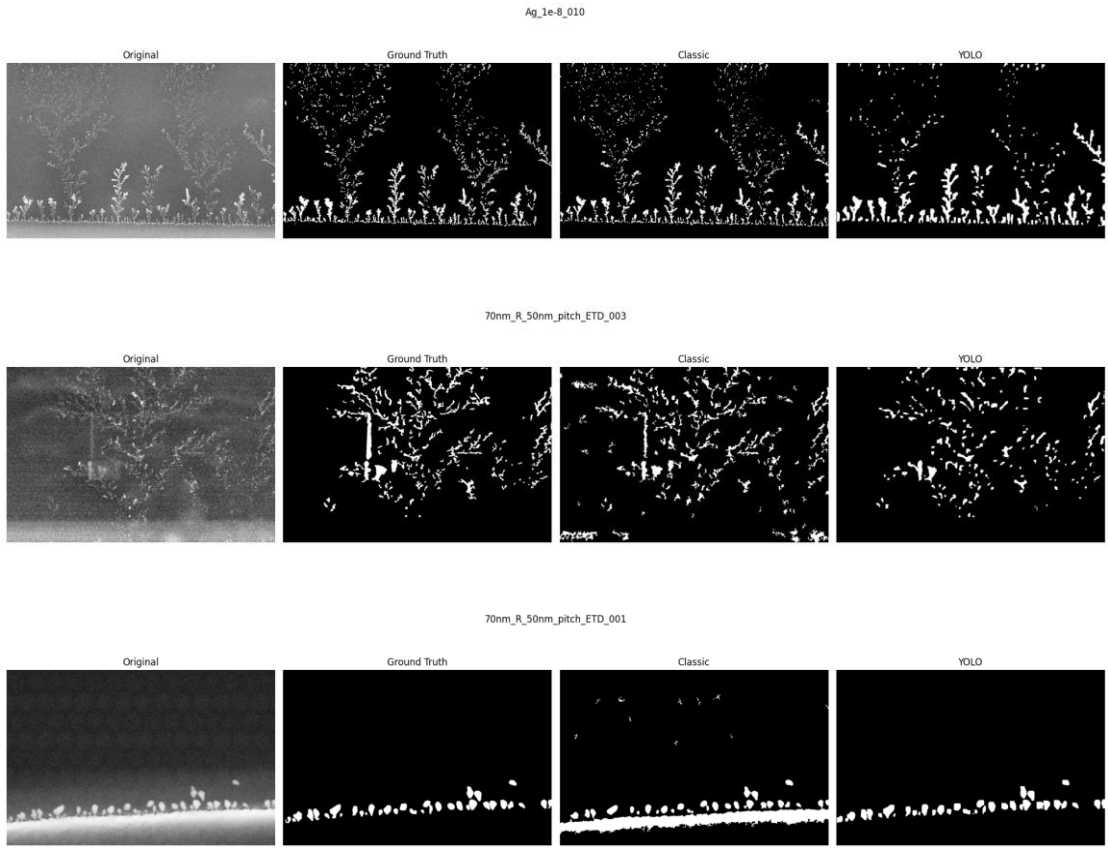
However, the classical pipeline performs competitively in several images and in some cases achieves **comparable segmentation quality**.

This variability suggests that **classical image processing techniques can still provide meaningful segmentation results**, especially when dendrite structures are clearly visible and separable from the background.

Qualitative Comparison

To further analyze the segmentation performance, several visual examples were examined.

The following example shows a comparison between the **original SEM image**, the **ground truth annotation**, the **classical segmentation**, and the **YOLO segmentation result**.



The visual comparison highlights the **structural differences between the two segmentation approaches**.

The classical pipeline tends to produce **cleaner and more conservative masks** that focus on the most prominent dendrite structures. Because it relies on deterministic morphological operations, it may sometimes miss **faint or fragmented dendrite branches**.

The YOLO segmentation model generally captures **a larger portion of the dendrite network** and tends to detect additional branches that may not be extracted by the classical pipeline. However, the YOLO model may occasionally introduce **slightly noisier regions or small over-segmented structures**.

Overall, the classical pipeline produces **more precise but conservative segmentations**, while the YOLO model provides **broader coverage of dendrite structures**.

Failure Analysis

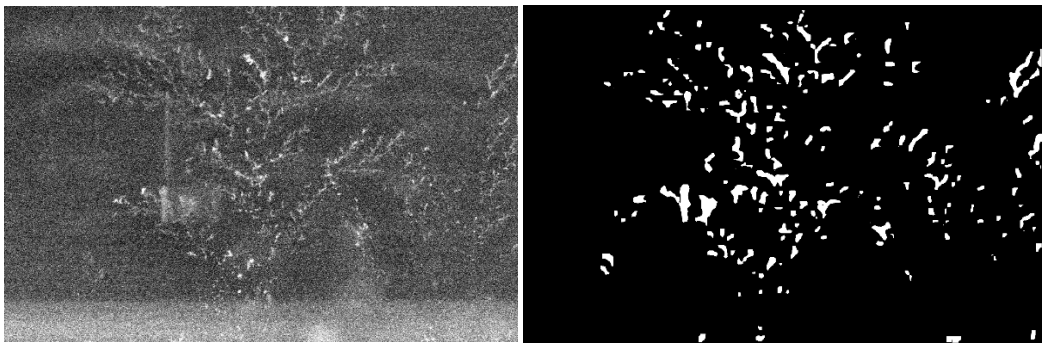
The classical image processing pipeline occasionally produces **false positives caused by SEM background texture and illumination artifacts**. In some images, the pipeline also fails to correctly separate thin dendrite branches from surrounding structures, leading to **fragmented or incomplete segmentation masks**.

In addition, the classical pipeline may miss **very faint dendrite branches** when their intensity is close to the background noise level.

The YOLO segmentation model generally produces **cleaner masks and captures a larger portion of the dendrite network**. However, the model may occasionally miss **extremely thin dendrite tips or fragmented branches**, particularly in the most complex HARD images.

These failure cases demonstrate the **complementary strengths and limitations** of the two approaches. While the classical method may be **more precise in some cases**, the deep learning model is generally **better at capturing complex dendrite morphology**.

Failure YOLO



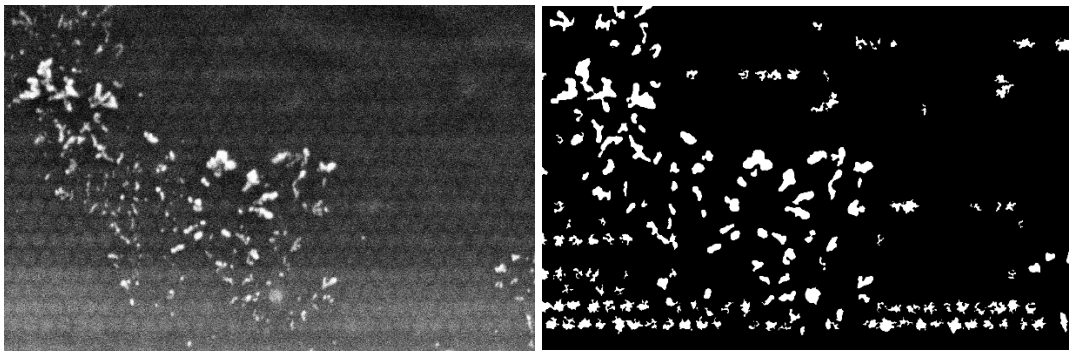
Example failure case of the YOLO segmentation model.

In this example, the YOLO model fails to detect several very thin dendrite branches present in the ground truth annotation.

Such thin structures may be difficult for the model to detect because they occupy only a very small number of pixels and may appear similar to background noise in SEM images.

Although the YOLO model generally captures the main dendrite structures, extremely thin tips or fragmented branches may occasionally remain undetected.

Failure Classical



Example failure case of the classical pipeline.

In this example, the classical image processing pipeline incorrectly detects background texture as dendrite structures.
This occurs because SEM images often contain granular noise and illumination artifacts that can resemble thin dendrite patterns.

Since the classical pipeline relies on intensity-based thresholding and morphological operations, it may incorrectly classify textured background regions as dendrites.

As a result, the segmentation mask contains several **false positive regions** that do not correspond to real dendrite structures.

5. Conclusion

This project compared a **classical image processing pipeline** with a **deep learning segmentation model (YOLO)** for detecting dendrites in **SEM images**.

The **classical pipeline** provides an interpretable **baseline method** and is able to capture a substantial portion of the dendrite structures. It achieves higher **Precision**, producing relatively conservative masks with fewer **false positive detections**. However, the method is sensitive to **background noise**, **illumination artifacts**, and **parameter tuning**.

The **YOLO-based segmentation model** achieves slightly better overall performance, with higher **IoU**, **Dice**, and **Recall** scores. This indicates that the model detects a larger portion of the dendrite structures and better handles **complex dendritic morphology**.

Overall, the results show that **deep learning** provides a modest improvement in **segmentation performance**, while the **classical pipeline** remains a competitive and interpretable baseline.

It is also important to note that the dataset used in this project is relatively small (**14 images**), which may limit the full potential of deep learning models. With larger datasets, the performance gap between **classical** and **learning-based approaches** may become more pronounced.

Future Work

Future work could include expanding the dataset with additional **SEM images**, applying **data augmentation techniques**, and training larger **segmentation models**.

Another possible direction is combining **classical preprocessing** with **deep learning segmentation models** to improve robustness under strong **SEM noise conditions**.